

# M/STAT 501 Fall 2025 Course Calendar

**Back to M/STAT 501 syllabus...** STAT 501 aims to cover Chapters 1–4 in Casella and Berger, continuing with Chapters 5–10 in STAT 502 next semester.

## Week 1: Aug 20–22

### *Wednesday*

- Introductions to the course and each other
- Sections 1.1–1.2.1: Set theory, sigma algebra, basics of probability theory

### *Friday*

- Section 1.2.2: Calculus of probabilities
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## Week 2: Aug 25–29

### *Monday*

- Section 1.2.2: Calculus of probabilities (cont)

### *Wednesday*

- Section 1.2.3–1.2.4: Combinatorics in probability
- **Homework 1 due** (Sections 1.1–1.2.2) in Gradescope by 5:00pm

### *Friday*

- Section 1.2.3–1.2.4: Combinatorics in probability (cont)
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## Week 3: Sep 1–5

### *Monday*

- *No classes: Labor Day*

### *Wednesday*

- Section 1.3: Conditional probability and independence
- **QUIZ 1:** Section 1.1–1.2.2

### *Friday*

- Section 1.3: Conditional probability and independence (cont)
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## Week 4: Sep 8–12

### *Monday*

- Section 1.3: Conditional probability and independence (cont)

### *Wednesday*

- Section 1.4: Random variables
- Section 1.5: Distribution functions
- **Homework 2 due** (Sections 1.2.3–1.3) in Gradescope by 5:00pm

### *Friday*

- Section 1.5: Distribution functions (cont)
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## Week 5: Sep 15–19

### *Monday*

- Section 1.6: Density and mass functions

### *Wednesday*

- Section 1.6: Density and mass functions (cont)
- **QUIZ 2**: Sections 1.2.3–1.3

### *Friday*

- Section 2.1: Distributions of functions of a random variable
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## Week 6: Sep 22–26

### *Monday*

- Section 2.1: Distributions of functions of a random variable (cont)

### *Wednesday*

- Section 2.2: Expected values
- **Homework 3 due** (Sections 1.4–1.6) in Gradescope by 5:00pm

### *Friday*

- Section 2.2: Expected values (cont)
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## Week 7: Sep 29–Oct 3

### *Monday*

- Section 2.3: Moments and moment generating functions

### *Wednesday*

- Section 2.3: Moments and moment generating functions (cont)
- Section 2.6: Cumulant generating functions and characteristic functions
- (Skip Section 2.4)
- **QUIZ 3**: Sections 1.4–1.6

### *Friday*

- Attend the Montana ASA Chapter Meeting in Wilson Hall 001 instead of class. Register and view the schedule here: <https://samidhasshetty.github.io/MT-ASA-2025/>
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## Week 8: Oct 6–10

### *Monday*

- Sections 3.1 and 3.2: Discrete families of distributions

### *Wednesday*

- Section 3.2 (cont): Discrete families of distributions
- **Homework 4 due** (Sections 2.1–2.3) in Gradescope by 5:00pm

### *Friday*

- Section 3.3: Continuous families of distributions
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## Week 9: Oct 13–17

### *Monday*

- Section 3.3 (cont): Continuous families of distributions

### *Wednesday*

- Section 3.4: Exponential families
- Section 3.5: Location and scale families
- Section 3.6: Inequalities and Identities

### *Friday*

- Section 4.1: Joint and marginal distributions
  - **QUIZ 4**: Sections 2.1–2.3
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## Week 10: Oct 20–24

### *Monday*

- Section 4.1 (cont): Joint and marginal distributions

### *Wednesday*

- Section 4.1 (cont): Joint and marginal distributions
- **Homework 5 due** (Sections 3.1–3.6) in Gradescope by 5:00pm

### *Friday*

- Section 4.1 (cont): Joint and marginal distributions
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## Week 11: Oct 27–31

### *Monday*

- Section 4.2: Conditional distributions and independence

### *Wednesday*

- Section 4.4: Hierarchical models and mixture distributions
- **QUIZ 5**: Sections 3.1–3.6

### *Friday*

- Section 4.4: Hierarchical models and mixture distributions (cont)
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## Week 12: Nov 3–7

### *Monday*

- Section 4.5: Covariance and correlation (bivariate)

### *Wednesday*

- Section 4.5: Covariance and correlation (cont)
- **Homework 6 due** (Sections 4.1–4.2, 4.4) in Gradescope by 5:00pm

### *Friday*

- Section 4.3: Bivariate transformations
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## Week 13: Nov 10–14

### *Monday*

- Section 4.3: Bivariate transformations (cont)

### *Wednesday*

- Section 4.6: Multivariate distributions
- **QUIZ 6**: Sections 4.1–4.2, 4.4

### *Friday*

- Section 4.6: Multivariate distributions (cont)
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## Week 14: Nov 17–21

### *Monday*

- Section 4.6: Multivariate distributions (cont)

### *Wednesday*

- Section 4.7: Cauchy-Schwarz and Jensen's inequalities
- Covariance and correlation for random vectors (not in textbook)
- **Homework 7 due** (Sections 4.3, 4.5–4.7) in Gradescope by 5:00pm

### *Friday*

- More on random vectors
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### *Nov 24–28: Fall break*

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## Week 15: Dec 1–5

### *Monday*

- More on random vectors

### *Wednesday*

- **Optional group review quiz** - option to work in teams during class; this score will replace your second lowest quiz score.

### *Friday*

- Final exam review
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## Finals week

- Final exam in Wilson Hall 1-124 on Monday, December 8, 10:00am-11:50am